

Noise Deleter: Adaptive Noise Cancellation Speaker

Final Report

ECE 499

August XX, 2026



University
of Victoria



Group ID: 17

Faculty supervisor: Peter Driessen

Co-supervisor (if any):

Team Information:

S. No.	Name	V Number
1	Sophie Gander	V00983673
2	Karnika Sitolay	V00955501
3	Nicholas Guildford	V00950036
4	Benjamin Blake	V00959199
5	Finn Berg	V00959800



Acknowledgments

The members of this project would like to acknowledge the following people for their support of this project:

- Dr. Peter Driessen for his advice on our project methodology
- Niloofar Tavahoodi for her project guidance throughout the term
- Asif Ali for his assistance throughout the term
- Dr. Michael McGuire and the Electrical Engineering Department for their financial support
- Dr. Sana Shuja for her facilitation of the ECE 499 class and assistance to our group
- JLC PCB for their timely manufacturing and delivery of our custom PCB
- The UVic Formula Racing team for loaning consumable supplies and testing equipment
- The family and friends of each team member who supported us during our capstone
- Brent Sirna for facilitating the ECE 499 lab space

Table of Contents

List of Figures	5
Executive Summary - KAr	5
1 Introduction	5
2 Objectives	6
3 Design Specifications	6
3.1 Software - Ben	7
3.1.1 Algorithms and Processing	7
3.1.2 Theory, Equations, and Research Basis	8
3.1.3 DSP Plan	9
3.2 Hardware - Nick	10
3.2.1 Microcontroller	10
3.2.2 Power	10
3.2.3 Microphones	10
3.2.4 Amplifiers	11
3.2.5 Speakers - ben help?	11
3.3 Enclosures - Finn	11
4 Literature Survey - Ben, Finn	12
4.1 Active Noise Control Fundamentals	12
4.2 Local vs Spatial Cancellation	12
4.3 Passive and Active Noise Barriers	13
4.4 ANC in the Built Environment and Project Justification	13
4.5 Comparison of Existing Solutions	14
5 Team Duties & Project Planning - Sophie	14
5.1 Encountered Challenges	18
5.2 Measures of Success	18
6 Design Methodology & Analysis	19
6.1 Software - Ben	20
6.2 Hardware - nick	21
6.3 Enclosure - Finn	21
7 Design & Prototype - Kar	22
8 Testing & Validation	23
8.1 Test Strategy	23
8.2 Hardware Validation	24
8.2.1 Breadboard Validation - kar Add test plan in appendix	24
8.2.2 PCB Validation (NICK pls confirm or add) + add your test plan to appendix	25
8.3 Software Validation (BEN pls confirm or add) + add your test plan to appendix	26
8.3.1 I ² S Integration Challenges	26
8.4 Full System Integration Testing	27
8.4.1 Adaptive Noise Cancellation Validation (BEN pls confirm or add)	27

8.5 Discussion & Future Improvements (PLS ADD AS U SEE FIT)	28
9 Cost Analysis - Nick, sophie will make table	28
10 Conclusion & Recommendations Finn	28
11 References BEN	29
Appendix A: Work Breakdown Structure	29

List of Figures

List of Tables

Executive Summary - KAr

An executive summary is a single page or single paragraph description of the work. The summary starts by highlighting the need for the work done. Then it moves on to describe in brief about the actual work with focus on the method used to accomplish the objective. The summary ends by giving an overview of the major outcomes of the work. It is to be noted that the summary will not contain any figures, references, equations or calculations. The purpose of the executive summary is to invoke enough interest for a reader to continue reading your actual report.

1 Introduction

The Noise Deleter suppresses noise levels in its environments using active spatial cancellation techniques [1]. This project was chosen because it catered to each team member's technical competencies and interests, including digital signal processing (DSP), hardware design, testing and validation, and the fabrication of an enclosure. Noise pollution in public and professional settings negatively affects quality of life for people with sensory issues and can cause hearing damage to workers at construction sites and in industrial environments such as generator facilities. This modular solution can be implemented in a variety of environments to:

- Support individuals with sensory issues
- Reduce exposure to harmful noise levels
- Provide targeted noise suppression

Potential users of the product include conscientious construction site managers, staff in special needs care facilities, and industrial facility managers. This project specifically aligns with the first guideline of the EGBG code of ethics: “hold paramount the safety, health, and welfare of the public, including the protection of the environment and the promotion of health and safety in the workplace” [2].

2 Objectives

The project objectives were established as follows:

- To cancel a 2kHz tone with a reduction of 15dB in a 2m range and 50cm x 50cm zone
- To prototype and fabricate a custom enclosure
- To design and assemble a fully custom PCB
- To ensure power consumption < 10W

3 Design Specifications

This section should highlight the technical ratings, and boundary conditions within which the design is expected to work. For example, if the project involves a filter design, the range of frequencies the filter is expected to work for has to be mentioned clearly in this section. There must be clear reasoning as to why the specified range is appropriate.

Please note, detailed design specifications for every major subsystem of the design has to be included. For example, if the project is related to a vacuum cleaner, then details about the power supply ratings, motor ratings, power converter ratings, mechanical casing etc. have to be furnished with clear justification.

As an engineering team, each group is expected to come out with details on how the components ratings were decided. Please note, in case equations have to be used please use equation editor and number the equations in sequence as shown below.

$$I_{dc} = \frac{V_{dc}}{R_1 + R_2} \quad (1)$$

Also, please do cite your references using numbers within square brackets [1], [2] and paraphrase from your reference articles. It is not acceptable to copy and paste content from references.

Relate to EGBC code of ethics when applicable. For example, if there are standards and/or regulations in the field of the project, please cite and discuss.

There are 4 essential systems of the adaptive noise cancellation project: software, firmware, hardware, and enclosures. Each system has its own unique design considerations and specifications in addition to the interdependencies between them.

3.1 Software - Ben

3.1.1 Algorithms and Processing

The cancellation engine runs an FxLMS adaptive filter, the standard approach for active noise control, dating back to Burgess's original adaptation of LMS for sound cancellation [9] and later formalized into the feedforward/feedback framework most ANC textbooks use today [10, 11]. The current target is a single 2 kHz tone, standing in for something like an HVAC blade-pass tone or fan whine. Restricting the problem to one known frequency simplifies things considerably: a general ANC system needs a full FIR filter dozens to hundreds of delayed-sample taps for every speaker, because it has to reproduce arbitrary noise, whereas a single known frequency can be fully described by one complex number per channel, whose real and imaginary parts together encode the gain and phase shift the controller should apply. So instead of a full filter per speaker, the system needs only one complex weight per speaker — eight numbers total instead of a multi-hundred-tap filter bank.

Once per audio sample, the firmware generates a reference signal at the target frequency, represented as a unit-length rotating vector in the complex plane (a phasor) that advances by a fixed small angle each sample ($X(n+1) = X(n) \cdot e^{j\omega_0}$) rather than recomputing cos/sin from scratch a meaningful CPU saving at 48,000 samples per second, with floating-point rounding drift corrected once every 64 samples by renormalizing the vector to unit length. Each speaker's output is then computed as $y_k(n) = \text{Re}\{W_k \cdot X(n)\}$, multiplying the speaker's complex weight by the current reference phasor and taking the real part. The two error-microphone samples $e_m(n)$, recorded near the listener's ears, are read next, and every speaker's weight is updated using the FxLMS rule: $W_k \leftarrow (1 - \lambda) \cdot W_k - \mu_{\text{eff}} \cdot \sum_m [e_m(n) \cdot \text{conj}(S_{km} \cdot X(n))]$ each weight moving a little in the direction that would have reduced that sample's error, scaled by how strongly the speaker couples to that microphone through the secondary path S_{km} .

The secondary paths S_{km} — one complex number per speaker-microphone pair, 16 total across 8 speakers and 2 microphones are measured once offline before deployment: a test tone is played through each speaker individually while the others stay silent, both microphones are recorded, and S_{km} is computed as $\text{DFT}(\text{mic}, f_0) / \text{DFT}(\text{reference}, f_0)$. Those 16 values are hard-coded into the firmware as a starting point. The system runs on an ESP32-S3 microcontroller, with real-time audio processing pinned to one of its two 240 MHz cores so the other stays free for WiFi/Bluetooth telemetry; the audio task runs at a slightly lower priority than the chip's internal inter-core messaging task, a deliberate choice that avoids a deadlock where a background flash

operation on one core could otherwise stall waiting on a handshake from the core that's busy processing audio.

3.1.2 Theory, Equations, and Research Basis

Several of the firmware's parameters reflect deliberate design choices rather than defaults. The step size is normalized rather than left as a raw value because, in plain LMS, the safe step size depends on how strong the secondary path is, something that isn't known until after the hardware has been measured. Dividing by the total secondary-path power ($\mu_{\text{eff}} = \mu / (\epsilon + \sum_k \sum_m |S_{km}|^2)$, with ϵ guarding against division by zero) turns the stability condition into a clean, hardware-independent rule: keep μ between 0 and 2, regardless of how strong or weak the real secondary paths turn out to be. A small leakage term λ (currently 0, i.e. off) is also built into the update rule, very slightly shrinking the weights each sample to prevent drift toward unbounded values if the controller's internal model the reference signal, the measured secondary paths doesn't perfectly match reality over a long run; this matters because the proof-of-concept's reference signal is generated internally rather than measured from a real microphone, so it's only an approximation of the true disturbance. Eight adjustable complex weights against only two error microphones also leaves the problem underdetermined there are more knobs than constraints, so a perfect solution exists in principle, and in practice the achievable cancellation is limited by how accurately the secondary paths were measured rather than by a shortage of speakers. All of this is verified offline, before any hardware is touched, using a Python simulator (anc_sim.py) that replays the exact update rule the firmware runs.

Two specific failure modes are checked quantitatively in that simulation before anything is flashed to hardware. If the disturbance sits at $f_0 + \Delta f$ while the internally generated reference stays fixed at f_0 , cancellation degrades as the mismatch grows, capping out around 20–25 dB of reduction at just a few Hz of drift. And if the measured secondary path S_{km} doesn't match the true path's phase — often because the measurement was taken through equipment with different timing than the deployed system the weight update starts pushing in the wrong direction, and past roughly a 90° phase error the system diverges outright instead of converging.

This approach follows an established, widely published line of work: narrowband tonal ANC using a feedforward/feedback structure has demonstrated 15–25 dB noise reductions in HVAC and fan-tone applications [10, 11], the choice to place the reference and error microphones in a headrest near the listener's ears follows a published active-headrest design [12], and the trade-offs around speaker count and spacing follow published work on car-cabin loudspeaker arrays for personal audio [13].

Narrowband cancellation was chosen over the original plan of covering a full 300 Hz – 4 kHz band because three separate calculations ruled the broadband approach out. First, causality: a cancelling system has to react within roughly a quarter-period of the disturbance to stay usefully in phase with it, and at 4 kHz that quarter-period is only about 62 microseconds for the entire

signal chain microphone, analog-to-digital conversion, computation, digital-to-analog conversion, and speaker an extremely tight budget given that a single sample at the 48 kHz sample rate already takes about 21 microseconds by itself. Second, the size of the "quiet zone": when cancelling sound in open air rather than at a microphone, the region where the cancelling wave and the noise actually line up shrinks roughly in proportion to the wavelength, about one-tenth of it, which at 4 kHz works out to only ~ 0.86 cm, smaller than the natural movement of a person's head while sitting. Third, speaker directionality: small speakers only project sound in a focused direction once the speaker itself is acoustically large compared to the wavelength, which for the drivers and array size in use only holds above roughly 700 Hz; below that, the array radiates almost equally in all directions, undermining the entire premise of a directional system in exactly the band where the first two problems would otherwise still allow cancellation. None of these three problems apply at 2 kHz with a real microphone providing the reference: the reaction-time budget relaxes to about 125 microseconds, the quiet-zone concern disappears once a real microphone defines the target point instead of open-air interference, and 2 kHz sits comfortably above the ~ 700 Hz threshold where the array stops being directional, which is the reasoning behind the current target frequency.

3.1.3 DSP Plan

In simple terms, the signal chain works as follows: an internally generated reference tone feeds the eight complex weights, which become eight separate speaker outputs sent over the microcontroller's digital audio bus to four DAC chips driving the eight speakers, while two microphones near the listener's ears feed back into the same per-sample update. Audio is processed in chunks of 64 samples at a 48 kHz sample rate (about 1.33 milliseconds per chunk), and if either the incoming microphone data or the outgoing speaker data stalls for more than 100 milliseconds, the firmware logs a warning rather than silently hanging.

Several things are being watched closely as the design matures. The most fragile parameter is the secondary path's measured phase: errors beyond roughly 90° flip the sign of the update and cause divergence, and because phase error scales directly with timing error ($\text{phase error} = 360^\circ \times f_0 \times \Delta t$), just 100 microseconds of unaccounted latency at the 2 kHz target easy to introduce by measuring through a PC sound card instead of the deployed hardware already works out to about 72° of phase error, dangerously close to that threshold; the fix is to always measure the secondary path through the real deployed signal chain, ideally on-device. Whether the internally generated reference stays in step with the real disturbance matters too, since that only holds true while the same board generates both the noise and the reference (as in this proof-of-concept) or once a real reference microphone is added; any drift between the two shows up directly as a ceiling on achievable cancellation, which can be predicted in simulation ahead of hardware testing. Because the step size is normalized by the measured secondary-path strength, its 0–2 stability range holds regardless of how strong or weak the real paths turn out to be, so stability doesn't hinge on guessing that scaling correctly though the leakage term, currently off, is there to

bound weight drift if the reference and disturbance fall out of step over a long run, and will be tuned once real running data is available. Live diagnostics of each microphone's error level and the magnitude of each speaker's weight, logged once a second, are what will confirm the system is actually converging and holding steady cancellation once it's running on real hardware, not just behaving well in simulation.

Every change to the step size, leakage, or secondary-path data is run through the Python simulator first, reproducing the exact update rule the firmware uses, including the two stress-test scenarios above. Catching an unstable configuration in simulation is far cheaper than discovering it acoustically on the bench. From here, the plan moves from a single tone (the current stage) to multiple simultaneous tones each with its own weight, then to a frequency that changes over time and is tracked automatically rather than fixed, and eventually to steering the cancellation beam to follow the listener's position rather than staying fixed.

3.2 Hardware - Nick

3.2.1 Microcontroller

An ESP32-S3 processor was chosen, packaged in a WROOM module for ease of testing and design. The controller includes built in USB OTG and I2S protocol, which are both very important for the design of the system. I2S protocol is used for the FxLMS software, microphones, and amplifiers. STM32 lacks this protocol, leading us to choose ESP32.

3.2.2 Power

Multiple buck converters will be used for the on-board power supply. The 12V-5V buck converter will need to output around 1W per amp and 1.5W 3V3 buck for the ESP32 chip. An emphasis will be made on maintaining clean power to achieve maximum fidelity in the mic and amplifier stages. To maintain a high quality 3V3 for the microphones, a separate LDO is used to convert the 5V into 3V3, eliminating noise issues from the buck solution.

Add all buck converter equations and calcs

Add all current flow calcs for the amps, to show how this design necessitated a buck converter and couldn't just be powered off the microcontroller

3.2.3 Microphones

Initially there will be one input microphone to detect the incoming sounds, and an error mic to provide feedback for the noise cancelling algorithm. The input microphone will be fixed on the PCB but the error mic will be attached to a wire for repositioning depending on where we decide

to place the listener. The INMP441 I2S MEMs microphone will be used, which has a built in A/D, modulator, and I2S converter on the chip. This simplifies the design immensely, and provides high quality audio, resistant to PCB design induced noise.

Add reasoning for the mics, like any calcs you did nick?

3.2.4 Amplifiers

The speaker array will be driven using MAX98357A Class-D amplifier modules. The MAX98357A was selected because it contains an integrated DAC and amplifier, allowing digital I²S audio from the ESP32-S3 to directly drive the speakers without requiring additional analog circuitry.

Initially, off-the-shelf evaluation boards will be used to validate communication between the ESP32 and speakers before integrating the design into a custom PCB. Currently the plan is for the final system to use eight or twelve amplifier channels, one for each speaker in the array, enabling independent control of each speaker for future beamforming and active noise cancellation algorithms.

Careful PCB layout and grounding practices will be used to minimize interference between the speaker amplifiers and microphone inputs.

Show the gain calcs like expected dB

3.2.5 Speakers - ben help?

Add in current calculations for this, show the impedance matching expected power, like cause they're 8 ohm they're not perfectly matched but its ok, cause they just use slightly different power.

3.3 Enclosures - Finn

To achieve controlled directional sound, the array utilizes eight Dayton Audio CE32A-4 drivers, each 32 mm in size, spaced 4.3 cm apart for a total array length of 25.6 cm. This physical configuration is critical for the system's directional effectiveness; at the 2 kHz target frequency, the wavelength is approximately 17.2 cm, meaning the array is roughly 1.5 wavelengths long. This results in a beam half-power width of approximately 34°, narrow enough to demonstrate a significant sound-level drop-off for off-axis listeners.

The choice of an eight-speaker array over a four-speaker alternative (which would double the beam width to 68°) ensures the directional effect is measurable and distinct. The decision to operate at 2 kHz, rather than a lower frequency, resolves a conflict between directionality and

array size: since these small drivers only become directional when "acoustically large" relative to the wavelength a threshold around 700 Hz for this geometry, 2 kHz provides the necessary frequency to ensure directionality while remaining within the timing constraints for effective phase cancellation.

Additionally, the narrowband simplification of the processing load to approximately 5 million multiply-accumulate operations per second provided enough computational headroom to support all eight channels without the need for additional complex hardware. The full set of eight drivers costs approximately \$40, balancing the project's low-cost requirement with the necessary density for beamforming. The physical design consists of a main panel enclosure housing the array, with a separate mounting structure for the feedback microphone placed at the intended listening position to ensure accurate error measurement.

Add calculations for the spacing of speakers and expected formed cancellation lobe from this

4 Literature Survey - Ben, Finn

Relate to EGBC code of ethics when applicable. For example, if there are standards and/or regulations in the field of the project, please cite and discuss.

4.1 Active Noise Control Fundamentals

The driving principle behind the Beat Downer dates to Lueg's 1936 patent on cancelling sound by emitting an inverted "anti-noise" wave [1]. Practical implementation came with Widrow et al's adaptive filtering using an LMS filter driven by a reference correlated with the noise [2]. A recurring theme across these methods is the secondary-path delay, the acoustic and electronic latency between the controller output and the error sensor which any cancellation scheme must compensate for to keep the anti-noise correctly phased. The Beat Downer's core challenge, phase coordinating anti-noise emission against measured system delay, is the same secondary-path problem, so this body of work is directly relevant regardless of which specific control algorithm is adopted.

4.2 Local vs Spatial Cancellation

Active noise control systems divide broadly into two regimes. Conventional local ANC minimizes the signal at a discrete error microphone, producing a compact quiet region centred on that sensor [3]. The size of this region is set by the wavelength of the controlled tone: cancellation holds within a radius of roughly one tenth of a wavelength of the error microphone

before the noise and anti-noise drift out of phase. At the Beat Downer's 2 kHz peak design frequency, where the wavelength is approximately 172 mm, this corresponds to a quiet region on the order of a few centimetres in radius, approximately the size of a person's head. The Beat Downer therefore targets local cancellation at a defined listening point rather than a continuous spatial zone.

Spatial ANC, by contrast, seeks to minimize acoustic power over a continuous region using multiple distributed loudspeakers, with recent kernel-interpolation methods achieving higher regional reduction than conventional multichannel control [4]. However, enlarging the quiet region to a continuous zone requires transducer spacing below half the controlled wavelength [4], which at 2 kHz would demand a spacing under 86 mm resulting in a high channel count. The current scope is to simply prove any amount of cancellation is possible.

4.3 Passive and Active Noise Barriers

The most common existing solution, the passive barrier, is effective at high frequencies but poor at low frequencies and bulky. Additionally, it only shields its shadow zone, since sound diffracts over the top edge as a virtual line source [5]. Active noise barriers (ANBs) add secondary sources at the edge to cancel this diffracted field [6]. The ANB literature provides relevant design rules: control is stable and effective when error-microphone spacing is below half the noise wavelength, and attenuation is greatest with secondary sources nearest the primary source [6] [7]. Reported ANB prototypes on stationary low-frequency tones have achieved peak insertion losses near 15 dB [7], consistent with the Beat Downer's 15 dB target. These results establish both the achievable attenuation for tonal noise and the spacing constraints that distinguish point control from zone control, motivating the Beat Downer's point-control approach at its chosen spacing.

4.4 ANC in the Built Environment and Project Justification

The closest application domain is active control of noise through open apertures, surveyed by Lam et al. [8] and demonstrated on full-sized open windows, work that confronts the same problem as the Beat Downer, cancelling noise propagating freely through an open region, and establishes that achievable attenuation is governed primarily by control-source count and position [8].

High-performance spatial and built-environment ANC systems [4], [8] achieve large quiet zones but require high channel counts, laboratory transducers, and computationally intensive algorithms, while passive barriers [5] and consumer headphones are cheap but limited, the former ineffective at low frequency and the latter restricted to a single listener at the ear. The Beat Downer occupies a practical middle ground: a low-cost, self-contained cancellation panel on commodity hardware (ESP32, integrated I²S class-D amplifiers, off-the-shelf speakers) under

10 W, delivering head-sized local cancellation of a 2 kHz tone at a defined listening point. This positions it as a proof-of-concept bridging academic ANC work and a practical, manufacturable device for construction perimeters, conference rooms, and sensory-sensitive users.

4.5 Comparison of Existing Solutions

Existing approaches to localized noise reduction fall into four main categories, summarized in Table 1 below. Each is evaluated on the three axes more relevant to the Beat Downers's scope: cancellation capability, cost, and ease of use.

Table 1: Comparison of Existing Solutions

Solution	Capability	Low-Freq Performance	Cost	Ease of Use	Limitations
Passive barriers/ acoustic foam	Locks direct path. Shadow path only.	Relatively poor. Requires mass	Low	High (One-time installation)	Ineffective at low frequency. Diffraction leaks over edge
ANC Headphones	Strong at the ear	Good	Low-Moderate	High	Protects one listener at a time. No spatial zone
Laboratory spatial ANC	Continuous quiet zone	Good	Very high	Low (Complex DSP, many channels)	High channel count. Lab transducers, heavy computation
Active noise barriers	Cancels edge diffraction in shadow	Good (Target band)	Moderate - High	Moderate	Tied to physical barriers structure. Placement sensitive
Beat downer	Spatial zone. Modular	Target band (low-freq tone)	Low	Moderate	Proof of concept. Limited active region.

5 Team Duties & Project Planning

To establish deliverables for each member, first a detailed work breakdown structure of the entire project was created. This is included in Appendix A. Following task establishment, Each team member's roles and deliverables were divided by system and role (Table 2).

Table 2: Member Roles, Systems, and Deliverables

Team Member	Role	Technical Systems	Deliverables
Sophie	Project Manager	Hardware	● Schedule of design and component orders ● Work breakdown of subsystems ● Track progress and identify bottlenecks ● Make final poster ● Split work between team members ● Speaker amplifier system: ○ Component specification ○ Feasibility review ● Test bench creation and bench testing with breadboard prototype ● PCB integration ● Bring up testing including firmware and hardware integration
Kar	Integration Lead	Hardware, Firmware, Software	● Overall task establishment of project ● Make final poster ● Speaker amplifier system: ○ Component specification ○ Firmware to drive speakers ○ Feasibility review ● Test bench creation and bench testing with breadboard prototype ● PCB integration into test bench ● Bring up testing including firmware and hardware integration ● Software algorithm assistance as needed ● Tuning of final parameters with final assembly
Ben	Software Lead	Software, Firmware	● Interface design ○ Programming interface ● Software architecture ○ Github setup ● DSP (digital signal processing) ○ Input tone analysis code ○ Algorithms for output tone generation and modulation ○ Output feedback mic performance metrics and active tone modification code
Nick	Hardware Lead	Hardware, Firmware	● Overall PCB design including microphone, speaker, microcontroller, and power supply circuits

			• Input microphone system ○ Component specification ○ Circuit design ○ Testing ○ PCB design ○ PCB assembly and integration ○ Test bench code ○ Firmware integration • Microcontroller system ○ Circuit design ○ Testing ○ PCB design ○ PCB assembly and integration ○ Test bench code ○ Firmware integration • Power supply ○ Component specification ○ Circuit design ○ Testing ○ PCB design ○ PCB assembly and integration • Bring up testing including firmware and hardware integration • Tuning of final parameters with final assembly
Finn	Physical Structure Lead	Enclosures	• Main structure: ○ Mounting locations for input microphones and speakers ○ PCB mounting ○ Wire routing ○ Fitment testing and adjustments • Manufacturing of all housing components • Assembly of all housing components

To ensure the project was completed smoothly, dependencies for each deliverable were established. The dependencies are summarized in Table 3.

Table 3: Deliverable Dependencies for each System

Technical System	Deliverable	Deliverable Designation	Dependancies
Hardware	Microphone circuit design	H-Mic	
	Speaker amplifier circuit design	H-Spe	

	Power supply circuit design	H-Pow	H-Spe, H-Mic
	Microcontroller circuit design	H-Con	
	Final PCB circuit integration and design	H-PCB	H-Spe, H-Mic, H-Pow, H-Con
	PCB assembly	H-Ass	H-PCB
Firmware	Speaker integration	F-Spe	H-Spe
	Microphone integration	F-Mic	H-Mic
	Programming interface	F-Int	
	User interfacing	F-Use	
Software	Overall test bench code	S-Tes	H-PCB
	DSP algorithms	S-DSP	
	Software architecture setup	S-Arc	
Enclosures	Main structure	E-Str	H-PCB
	Feedback microphone structure	E-Mic	H-Mic
	Manufacturing	E-Man	E-Str, E-Mic
	Assembly	E-Ass	E-Man

Based on how frequently deliverables appeared as dependencies in the above table, the obvious bottlenecks for the entire project were identified to be the hardware system, particularly the H-PCB deliverable. Several deliverables depended on the completion of the H-PCB deliverable as follows:

- **H-Ass (PCB assembly):** While the PCB was ordered with all circuit components already attached from the manufacturer, all off-board components like the microphone, speakers, and all wires had to be assembled by the team, and depended on the arrival of the finished PCB.
- **S-Tes (Overall test bench code):** The test bench code for the PCB was developed before the PCB design was finalized, but only when the final PCB design was made could it account for all peripherals and number of speakers, amplifiers, microphones...etc.
- **E-Str (Main structure):** The main structure houses the microphones, speakers, PCB, and all attached wires, and could not be designed until the final dimensions of the PCB, PCB components, and wire routing plan were known.

Because all circuits are on the same PCB, the H-PCB deliverable also depended on completing the design of all individual circuits: H-Mic, H-Spe, H-Con, and H-Pow. Essentially, the hardware system was a compounding bottleneck. The software and firmware systems only required basic hardware circuits to be completed before they could be almost fully designed, so they could be worked on in tandem. The enclosure was also preliminarily designed in parallel with the other systems. The potential for the hardware system becoming a significant bottleneck, especially with the goal of designing a fully custom PCB, was the main driver for assigning 3 team members (Nick, Kar, and Sophie) to help with hardware. This strategy was effective at mitigating delays, though with such a short time frame, the complexity of the PCB did delay testing.

5.1 Encountered Challenges

Overall, the main challenge of the project was planning the testing of the final system. The decision to design a fully custom PCB meant the final code could not be accurately tested before the PCB arrived. For any software heavy project like this one, tuning and code integration is inevitably the hardest and most time consuming task. The short time frame of this project hampered the team's ability to fine tune the end product. While the PCB did arrive with adequate time for assembly and testing, more testing time would have been optimal for fine tuning the systems performance.

5.2 Measures of Success

Variable measures of success were established for the project. It was anticipated that the complex DSP algorithms may be difficult to interface with the physical microphone and speaker circuits over the short testing period once the PCB arrived. Possible issues included processing speed between input and output microphones (phase coordination) and quality of tone input and generation. These milestones were designed to be sequentially achieved in order of difficulty to ensure the project could still be demonstrated and used as a proof-of-concept:

1. Measure the input tone, and play a tone at the same amplitude as the input tone
 - Proves interfacing of all system peripherals
 - Proves basic DSP algorithm functions
2. Cancel a single input tone
 - Proves destructive waveform generation
3. Cancel a sequence of predictable input tones
 - Proves active destructive waveform generation
4. Cancel a complex input

- Proves waveform generation algorithm is functioning exactly as desired, with the timing capabilities to cancel complex input signals such as loud construction noises or music

6 Design Methodology & Analysis

In this section, groups are expected to provide details of the design method adopted for the project. In case a couple of approaches were followed, each one should be elaborated. Further, other design alternatives explored by the group must be described in detail. The reason behind selecting the approach of choice has to be included. The section should also comprise of analysis for the method adopted. Analysis can be done using a set of mathematical equations, and/or using computer simulation tools. Results of the analysis backed by theoretical explanations are expected.

As a part of the report, if figures have to be included, it is preferable each graph/image/chart uses the template of the sample graph shown in Fig. 1. That is, the axes must be clearly labelled. It is preferable to have a legend. Font size should be readable in print version of the report. And a caption must be provided explaining what the image is about.

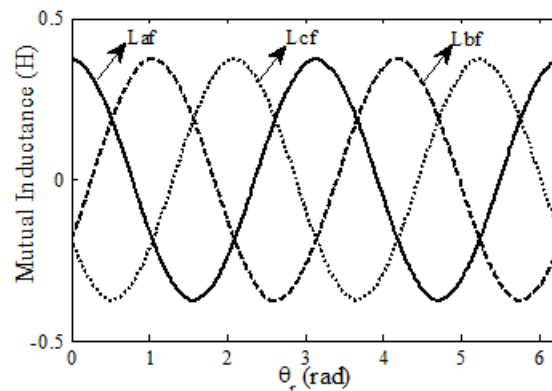
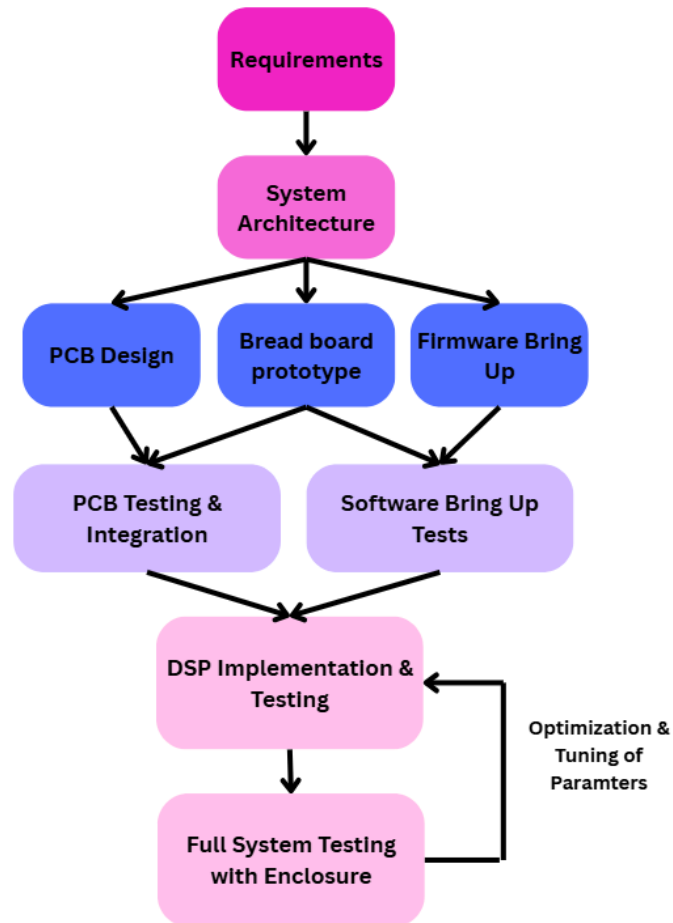


Fig. 1. A sample graph with labels along x, and y axes. The image has multiple signals represented using different types of lines and identified using a text.

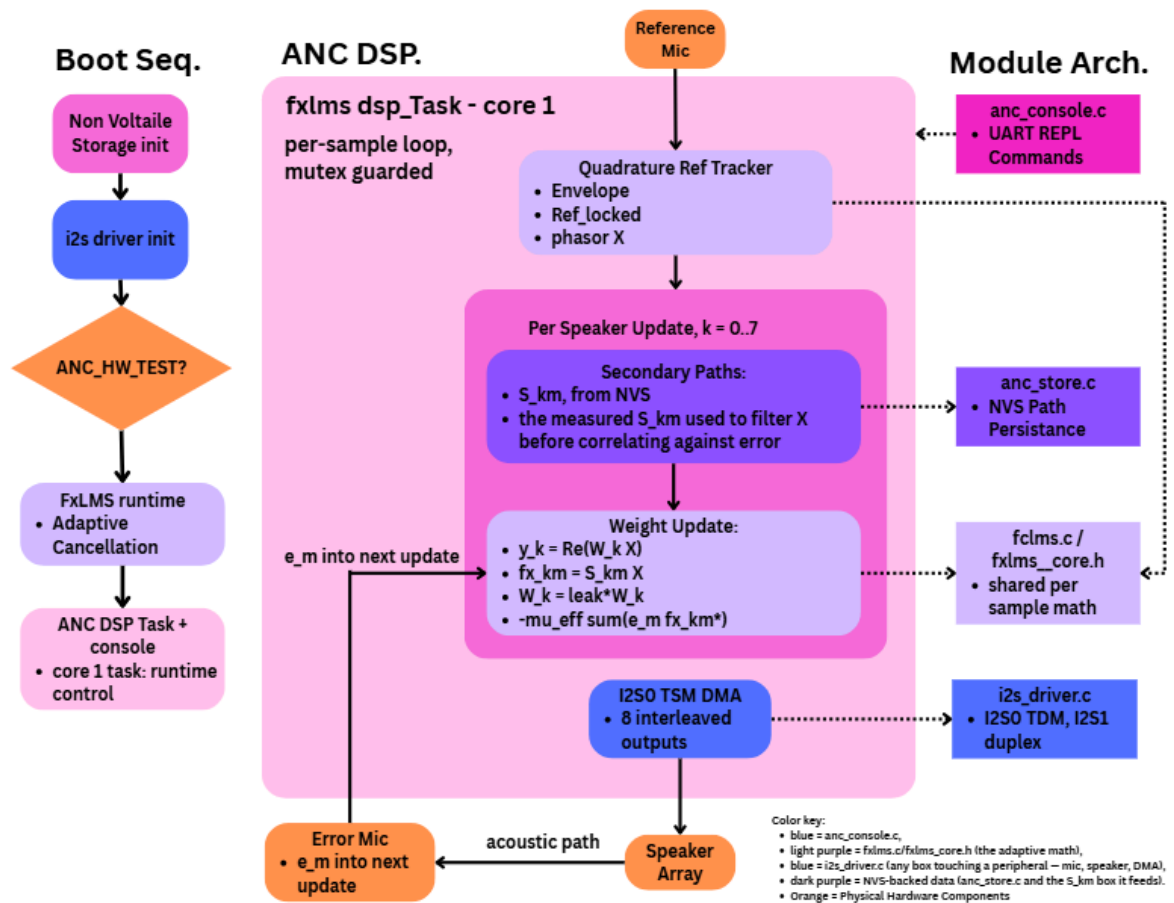
Relate to EGBC code of ethics when applicable. For example, if there are standards, policies and/or regulations in the field of the project, discuss how the design follows the standards/regulations, or not applicable. If the design methodology distinguishes facts from assumptions and opinions, clearly states the assumptions, etc.

The Noise Deleter was developed using an iterative systems engineering approach. Initial work focused on validating individual hardware subsystems in parallel with the firmware setup including drivers and libraries. This was followed by software implementation, hardware integration, and adaptive control development. Since active noise cancellation depends on both the physical acoustic system and the digital control algorithm, each subsystem was verified independently before integrating the complete closed-loop system. The following figure demonstrates development methodology.



6.1 Software - Ben

Add screenshots of parts of the code that are particularly important as figures and justification
Add justification for design for software architecture, file structure, rtos, etc.



6.2 Hardware - nick

Add reasoning for the pcb, design of power domains, reverse polarity, ldos, etc. Design review, Maybe split into like the different circuits and include all you altium schematics and then the overall board at the end, having an overall board section that also talks about trace and via design to minimize interference and such.

Add screens shots of schematic or put in appendix

Made block diagram for you already:

Can be like “this block diagram outlines the required functionalities, then the PCB was designed around the design specs” or however you want to put it

6.3 Enclosure - Finn

Add design iterations and all that for enclosure

Add 3d model screenshots

7 Design & Prototype - Kar

The final design of the final working prototype that meets all objectives have to be included in this section. Explanations are expected from the groups regarding how each objective is addressed by the final design. In case, some objectives were not met, reasons for the same have to be provided. Please note, it is acceptable to develop a working prototype that does not meet all the objectives. However, clarity on why certain objectives were not accomplished (example: feasibility, cost, time, priority etc.) should buttress your decision. Finally, explicit mention on the boundary conditions of the designed prototype must be provided. For example, if a power converter is built, specification on the input it can take, output it can work, maximum efficiency, duration of operation etc. has to be included.'

Explicitly state how each was satisfied or not and why. Also go reference objectives in the objectives section .

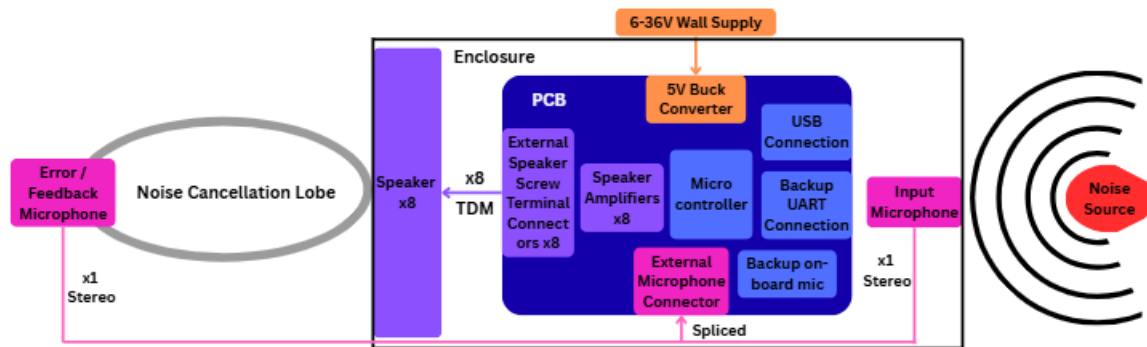
The final Noise Deleter prototype consists of a compact active noise cancellation system integrating custom hardware, embedded firmware, digital signal processing, and a 3D-printed enclosure into a single self-contained unit, as shown in the following figure. The system is designed to demonstrate localized cancellation of a tonal noise source by generating an adaptive anti-noise signal through an eight-speaker array.

The prototype is powered from a 6–36 V DC supply, which is regulated on-board using a 5 V buck converter to power the speaker amplifiers and a dedicated low-noise 3.3 V supply for the ESP32-S3 microcontroller and digital MEMS microphones. The ESP32-S3 performs all real-time digital signal processing, including microphone acquisition, adaptive filter updates, and generation of the anti-noise signals.

Eight Dayton Audio CE32A-4 speakers are individually driven by MAX98357A Class-D I²S amplifier channels operating in Time Division Multiplexing (TDM) mode, allowing synchronized control of each loudspeaker while minimizing wiring complexity. Acoustic feedback is provided by a reference microphone positioned near the disturbance source and an error microphone located at the desired listening position. Both microphones communicate digitally with the ESP32-S3 using the I²S interface.

The adaptive FxLMS controller calculates the required phase and amplitude corrections and distributes independent audio signals to each speaker through the TDM audio interface. The resulting sound field is monitored by the error microphone, which continuously provides feedback to update the adaptive controller and minimize the residual acoustic error.

The final hardware integrates the power supply, microcontroller, microphones, amplifier interfaces, speaker connectors, USB programming interface, and external microphone connector onto a single custom PCB enclosed within a compact housing.



Add image of final assembly

Explain how we met or didn't meet objectives and why

8 Testing & Validation

This section comprises of test plans adopted to test different features of your design and prototype. Any test plan should clearly detail the operating conditions (input applied, temperature, processor capability etc.) under which the test has to be carried out. Outcomes of test plan should be compared with expectations. Any deviation has to be justified. Typically, for hardware based prototypes, a circuit diagram is supplied in a test plan. In case of software based prototype, a block diagram or a flow chart is provided along with the test plan. **Note, this section will contain all the results obtained using the prototype.**

Test plan must be provided for all major subsystems of the prototype developed. A test plan must be repeatable from any geography as long as operating conditions are maintained and hence has legal implications.

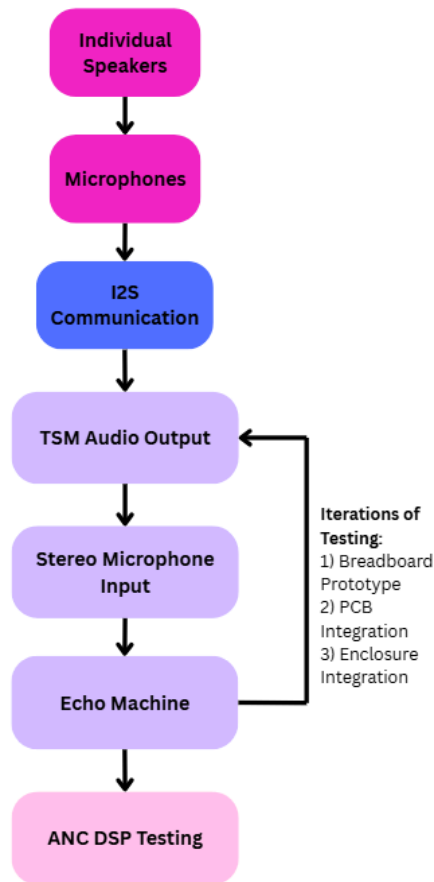
Validation is the process of running several test sequences as per test plans and demonstrating the ability of the prototype to meet the desired objectives. There can be situations where only 8 out of 10 objectives could be validated while 2 might not meet satisfactory performance requirement. Under such situations, these tests have to be marked as fail. Decision to accept a prototype as functional is possible even with failed test cases.

Relate to EGBC code of ethics when applicable. For example, if there are standards and/or regulations in the field of the project, please cite and discuss.

8.1 Test Strategy

Before attempting active noise cancellation, each subsystem was independently validated to ensure the hardware and firmware functioned correctly before closed-loop operation. Testing progressed from individual peripheral verification to complete system integration. This staged approach reduced

debugging complexity and ensured faults could be isolated before introducing the adaptive DSP algorithm. Figure X summarizes the validation process.



8.2 Hardware Validation

Hardware validation was broken up into prototyping on a breadboard using modular components to prove the circuit in design would be functional as well as functional testing of the PCB.

8.2.1 Breadboard Validation - [kar Add test plan in appendix](#)

Initial testing was performed using a breadboard prototype consisting of an ESP32-S3 development board, MAX98357A amplifier development boards, I²S MEMS microphone evaluation boards, and Dayton Audio speakers. This configuration allowed hardware and firmware to be verified before manufacturing the custom PCB.

The following functional tests were completed:

Test	Purpose	Result
Single speaker output	Verify amplifier communication	Pass
Sequential speaker tone generation	Confirm speaker wiring and channel mapping	Pass
Microphone input	Verify I ² S microphone operation	Pass
TDM amplifier output	Validate simultaneous multi-speaker audio	Pass
Stereo microphone configuration	Validate left/right microphone addressing	Pass
Echo machine	Verify complete input-to-output audio path	Pass

8.2.2 PCB Validation (NICK pls confirm or add) + add your test plan to appendix

Following PCB assembly, the same validation procedure used during breadboard development was repeated to verify that no functionality had been lost during hardware integration.

The following tests were performed:

- PCB power rail verification
- ESP32 programming and USB communication
- On-board MEMS microphone functionality
- Speaker amplifier operation
- I²S communication
- TDM speaker output
- Echo machine verification

Repeating the original bring-up tests ensured the PCB behaved identically to the development hardware before attempting adaptive noise cancellation.

8.3 Software Validation (BEN pls confirm or add) + add your test plan to appendix

Software validation was performed using an incremental integration strategy to verify the functionality of each firmware module before enabling the adaptive noise cancellation algorithm. Testing began with offline verification of the DSP algorithms using a Python simulation environment to confirm algorithm stability and predict expected cancellation performance prior to deployment on the ESP32-S3. Once verified, firmware was compiled and flashed onto the hardware prototype for real-time testing.

Initial firmware validation focused on low-level peripheral functionality. Individual speaker channels were exercised using a hardware check mode that sequentially generated test tones on each output channel, confirming correct TDM slot assignment and speaker mapping. Similarly, the reference and error microphones were verified independently by monitoring received audio data and confirming the correct left/right stereo channel configuration. These tests ensured that the digital audio interfaces were operating correctly before closed-loop testing commenced.

Following peripheral validation, a software loopback ("echo machine") test was performed in which microphone input was immediately reproduced through the speaker array. This test verified the complete digital audio pipeline, including microphone acquisition, DMA transfers, audio buffering, I²S communication, and speaker output. Successful completion of this test demonstrated that the firmware could reliably process real-time audio with minimal latency before introducing adaptive control.

The adaptive FxLMS controller was then evaluated using an externally generated tonal disturbance. The firmware was configured to measure the secondary acoustic paths, initialize the adaptive filter, and iteratively adjust the speaker outputs to minimize the measured error microphone signal. Throughout testing, diagnostic telemetry including adaptive filter weights, error microphone RMS levels, reference tracking status, convergence behaviour, and divergence protection were continuously monitored to evaluate controller stability.

Software robustness was further assessed through several stress tests. These included intentional frequency offsets between the disturbance and reference signal, adaptive gain variations, temporary loss of the reference signal, and forced divergence by increasing the adaptation step size. The controller successfully detected unstable operating conditions, muted the speaker array when required, and recovered once nominal operating conditions were restored. These tests verified the firmware's fault detection and protection mechanisms in addition to its nominal active noise cancellation performance.

The complete software validation procedure and detailed pass/fail criteria are documented in the project test plan provided in Appendix X.

8.3.1 I²S Integration Challenges

The largest integration challenge encountered during development involved the microphone interface.

The original architecture called for two error microphones and one reference microphone. However, the selected MEMS microphones only support stereo I²S operation, while the ESP32-S3 provides two I²S peripherals. Once one peripheral was allocated to the eight-channel TDM speaker output, only a single stereo microphone interface remained available. As a result, only one reference microphone and one error microphone could be connected simultaneously.

This limitation required the final prototype to operate using one reference microphone and one error microphone configured as the left and right channels of a stereo interface. Although this configuration successfully demonstrated active noise cancellation, it reduced the available spatial error information compared to the original design.

Future revisions should consider either:

- replacing the microphones with TDM-capable digital microphones, allowing multiple microphones to share a single audio bus, or
- selecting a processor that provides additional I²S peripherals.

8.4 Full System Integration Testing

After PCB validation, all hardware was installed into the final enclosure. Functional testing was again repeated to verify operation under final assembly conditions.

During integration, several hardware issues were identified.

The microphone wiring exhibited intermittent noise pickup. The exact root cause could not be conclusively determined and is currently under investigation. Possible causes include insufficient shielding, electromagnetic interference from nearby digital electronics, or antenna effects introduced by the microphone wiring. The microphone wiring was re-soldered and rerouted, which reduced the observed interference, although further root-cause analysis is required.

Several connector interfaces also proved unreliable during repeated assembly and disassembly. Some intermittent failures were traced to solder joints or mechanically weak connector interfaces. Future revisions should use more robust locking connectors and improved strain relief to improve long-term reliability.

8.4.1 Adaptive Noise Cancellation Validation (BEN pls confirm or add)

Once the complete system had been validated, testing progressed to the adaptive FxLMS controller.

Initial testing produced large oscillations and high-amplitude squealing noises, indicating instability within the adaptive feedback loop. Multiple software parameters including adaptive gain, initialization routines, and secondary-path calibration were iteratively refined until stable operation was achieved.

After tuning, the system demonstrated measurable adaptive behaviour. The error microphone response showed that the controller was generating an anti-noise signal and reducing the measured acoustic error,

indicating successful operation of the FxLMS algorithm. Although complete cancellation was not achieved, noticeable attenuation of the disturbance was observed.

Several factors limited overall cancellation performance:

- inaccuracies in secondary-path estimation,
- reflections within the test environment,
- acoustic coupling between the speakers and microphones,
- sensitivity to the exact disturbance frequency.

These factors are expected for a proof-of-concept operating in an uncontrolled acoustic environment.

8.5 Discussion & Future Improvements (PLS ADD AS U SEE FIT)

The prototype successfully demonstrated the complete active noise cancellation signal chain, from acoustic sensing through digital signal processing to anti-noise generation. However, the current implementation remains a narrowband proof-of-concept.

The controller assumes the disturbance frequency is known in advance and internally generates the corresponding reference signal. Consequently, the system is only capable of cancelling tones whose frequency matches the programmed reference.

The next stage of development is to implement automatic frequency estimation using the reference microphone. Rather than assuming a fixed 2 kHz disturbance, the DSP would continuously identify the dominant frequency and update the adaptive controller in real time. Extending this approach to track multiple simultaneous frequency components would allow the controller to generate several adaptive anti-noise signals concurrently, transforming the current proof-of-concept into a true adaptive active noise cancellation system capable of suppressing realistic broadband disturbances.

9 Cost Analysis - Nick, sophie will make table

The section should include both direct costs and indirect costs. Direct costs are money spent on products, travel, food etc. Indirect costs are money equivalent to expenses such as hours worked, overhead due to delay, consultation time & fees (if associated), overtime hours etc. Entire team's time has to be accommodated.

Cost analysis should also include values of funding as and where applicable. A tentative pricing and return on investment calculation has to be presented.

Finally, means to reduce the cost of the design/prototype need to be suggested.

Direct Costs			
Item	Quantity	Price (CAD)	Total Cost (CAD)
Test bench wires and breadboard	1	26.87	26.87
ECE Department Funding	1	-200	-200
Total Direct Costs:			
Labour Expense			
Item	Hours	Price (CAD)	Total Cost (CAD)
Project Management	10	29	
Documentation and Reporting	16	29	
Integration and Testing	16	29	
Software Design		29	
Hardware Design	4	29	
Physical Structure Design		29	
Total Labour Expenses:			
Total Project Costs:			

10 Conclusion & Recommendations Finn

In this section of the report, the groups are expected to provide clear conclusions based on the project. Conclusions can include statements affirming the accomplishment of set objectives. Conclusions can also have statements that bring out the limitations or challenges of the prototype.

Finally, future recommendations for the reader have to be presented.

Reference the measures of success in the project planning section and how far we got, then what would be needed to finish the entire working prototype

11 References BEN

[1] B. Lam et al., "Ten questions concerning active noise control in the built environment," Building and Environment, vol. 200, art. 107928, 2021.

[2] Engineers and Geoscientists British Columbia, *Code of Ethics*. Burnaby, BC, Canada. [Online]. Available: <https://www.egbc.ca/complaints-discipline/code-of-ethics/code-of-ethics> . [Accessed: July 26, 2026].

References cited should be easily obtainable. Please refrain from using sources that are not trust worthy. Preferable to refer text books and works published by international organizations of reput. It is mandatory to cite the source from which you learnt about a concept/idea. It is not acceptable to copy and paste images. They have to be re-drawn.

[1]

[2]

Appendix

1. Code of ethics of EGBC
2. Website link and the main page image
3. Details of data, test results, codes, etc, if applicable

Appendix A: Work Breakdown Structure

Major Task	Subtask	Description	Notes	Completed (Y/N)
Hardware				
Speaker and Noise source	Select noise source	Choose speaker capable of monotone output	personal speaker - should keep same one throughout	Yes
	Select speaker	choose speaker capable of range of outputs at volume to cancel		Yes
	Test tone generation	Verify speaker can produce stable tones	set a specific tone and volume as our initial endpoint for filtering, low medium volume? Write this	Yes

			down so we can reproduce	
Microphone	Select microphone	Choose mic with suitable voltage range and SPL handling		Yes
	Directional design	Evaluate unidirectional vs omnidirectional mic	array vs one	Yes
	Power requirements	Determine powering method for mic		Yes
MCU	MCU Selection	Select MCU capable of all tasks	ESP32 capability? (might need stm32) - no. adc -no. dac - clocking speed - floating point stuff - adc accuracy (12-bit?) stm32 has nice dsp library with fft	Yes
Signal Conditioning	Amplifier design	Design or select op-amp circuit		Yes
	Biasing circuit	Shift signal to ADC input range		Yes
	Anti-alias filter	Design low-pass filter before ADC		Yes
ADC Interface	Select ADC	Internal MCU ADC or external ADC decision		Yes
	Voltage compatibility	Ensure signal fits ADC range		Yes
Output Stage	DAC selection	Choose DAC or I2S interface		Yes
	Amplifier selection	Choose audio amplifier		Yes
	Speaker selection	Select output speaker	needs to be capable of same volume as speaker	Yes
Power Supply	Power architecture	Define system power sources		Yes
	Voltage regulation	Design regulators or dividers		Yes
Housing	Enclosure	Design enclosure in CAD		Yes

	design			
	Mounting PCB	Design mounting method		Yes
	Microphone Mounting	Design mounting method	- is it mounted on same thing as speaker bar? - is it in its own housing?	Yes
	Speaker mounting	Design speaker mounts		Yes
Validation	SPL measurement setup	Set up sound level meter testing		Yes
Firmware				
Audio Capture	ADC configuration	Set sampling rate and resolution		Yes
	DMA setup	Implement DMA buffering for ADC		Yes
	Buffer management	Implement circular buffers		Yes
DSP	FFT implementation	Implement FFT for frequency detection		Yes
	Phase estimation	Calculate phase of incoming signal		Yes
	Anti-noise generation	Generate inverted waveform		Yes
	Gain control	Adjust amplitude dynamically		Yes
Timing	Latency measurement	Measure system delay		Yes
	Synchronization	Align anti-noise phase timing		Yes
RTOS	Task scheduling	Decide RTOS or bare metal scheduling		Yes
	Interrupt handling	Handle ADC/DAC interrupts		Yes
	DAC output config	Configure DAC/I2S output		Yes

Output

	Audio streaming	Stream anti-noise signal continuously		Yes
Software				
Control Interface	Laptop interface	Design control via PC		Yes
	User controls	Implement buttons or UI		Yes
Testing	Test scripts	Write scripts for tone testing		Yes
	Data logging	Log audio data for debugging		Yes
Analysis	Frequency analysis tools	Develop visualization tools		Yes
	Performance metrics	Measure noise reduction effectiveness		Yes